

Buyu Deng

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🌐 <https://buyudeng.github.io/>

🐙 BuyuDeng



Education

2023.09 – Expected **B.S. Candidate**, *School of the Gifted Young*, University of Science and Technology of China (USTC)
2027.06 **Major:** Geophysics

GPA: 3.62/4.30 (**Rank:** 3/20)

Research Interests: Artificial Intelligence, Subsurface Energy Systems, representation learning and generative models for 3D geophysical data, geomodeling, and Forward Modeling

Research Experience

2026.07 – Present **Undergraduate Research Intern**, [Stanford Center for Earth Resources Forecasting \(SCERF\)](#), Stanford University
Supervisor: **Prof. Tapan Mukerji**

- Conduct research on geomodeling and Forward Modeling for subsurface energy systems.

2024.08 – Present **Undergraduate Researcher**, [Computational Interpretation Group](#)
Supervisor: **Prof. Xinming Wu**

3D "All-in-One" Seismic Denoising

- Built a synthetic 3D seismic dataset with random noise, acquisition footprints, and migration artifacts; trained an all-in-one denoising network based on Degradation Representation Learning.
Code: github.com/BuyuDeng/NDR_Seismic

Geo-VAE for 3D Geophysical Data Compression & Latent Processing

- Adapted Wan-VAE into a unified Geo-VAE for seismic volumes, RGT cubes, property models, structures, and geobodies, achieving $8 \times 8 \times 8$ volumetric compression with high-fidelity reconstruction.
- Enabled memory-efficient layer-by-layer inference on large 3D volumes using causal 3D convolutions and feature caching, avoiding chunk-boundary artifacts.
- Built latent-space downstream workflows for missing-trace interpolation with a dual-mask latent diffusion model and seismic denoising with an HRNet-style latent mapper.
Submission: full paper submitted to *Geophysics*; gave an oral presentation at IMAGE 2026.

Code: github.com/BuyuDeng/Geo-VAE

PHY-to-BGC Ocean Biogeochemical Emulation

- Built a 2.5D full-image U-Net emulator mapping global physical ocean fields to coupled BGC variables. The model uses monthly anomaly learning, seasonal/latitudinal auxiliary channels, longitude-circular padding, and global/vertical/basin-scale evaluation.
- Demonstrated that a model trained on normal years maintains strong performance in an El Nino year, suggesting robust generalization under anomalous ocean-climate conditions; **manuscript in preparation**.

GNSS-NISAR Ionospheric Super-Resolution Reconstruction

- Developed an automated pipeline to invert high-resolution ionospheric phase delays from NISAR LA/LB dual-frequency InSAR data, convert GNSS TEC differences into L-band ionospheric phase, and align both products in radar/geographic coordinates.
- Next step: train a super-resolution model using GNSS ionospheric phase as the low-resolution background and NISAR-derived phase fields as high-resolution supervision.

Honors and Awards

- 2025 **Zhao Jiuzhang Elite Program Scholarship**, School of Earth and Space Sciences
- 2024 **Rose Fund Endeavor Scholarship**, School of the Gifted Young
- 2024 **Zhao Jiuzhang Elite Program Scholarship**, School of Earth and Space Sciences
- 2023 **Freshman Scholarship**, University of Science and Technology of China (USTC)

Experience and Leadership

2025.08 **IPGP Geophysical Field Camp:** conducted geophysical surveys and joined academic seminars with faculty from the Institut de Physique du Globe de Paris (IPGP).

2025.06 – 2026.01 **Head (Minister)**, Dept. of Cultural & Sports in College Student Union